University of Cape Town



Machine Learning Assignment 1

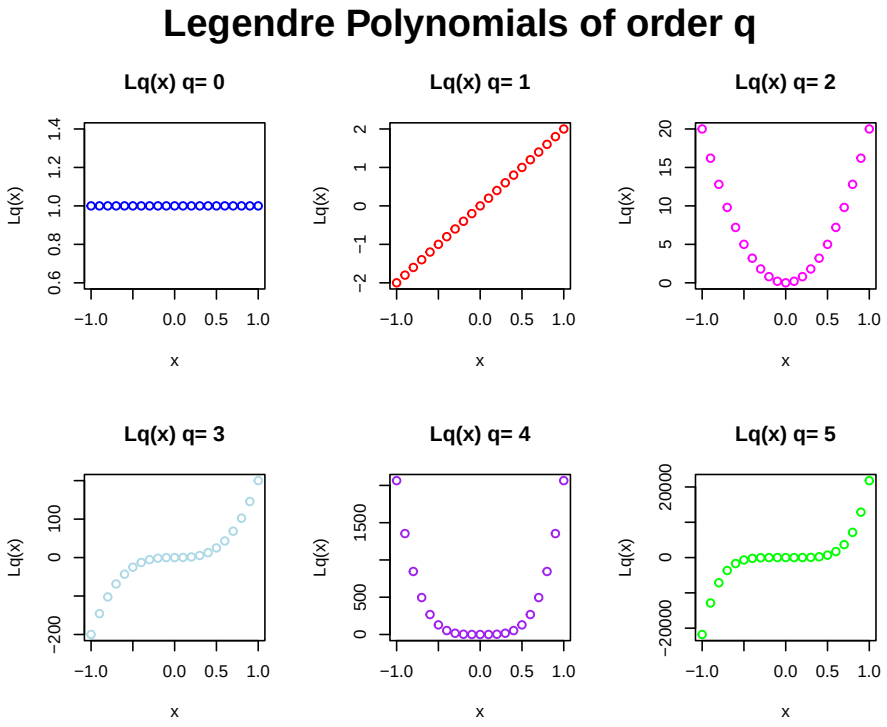
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Question 1

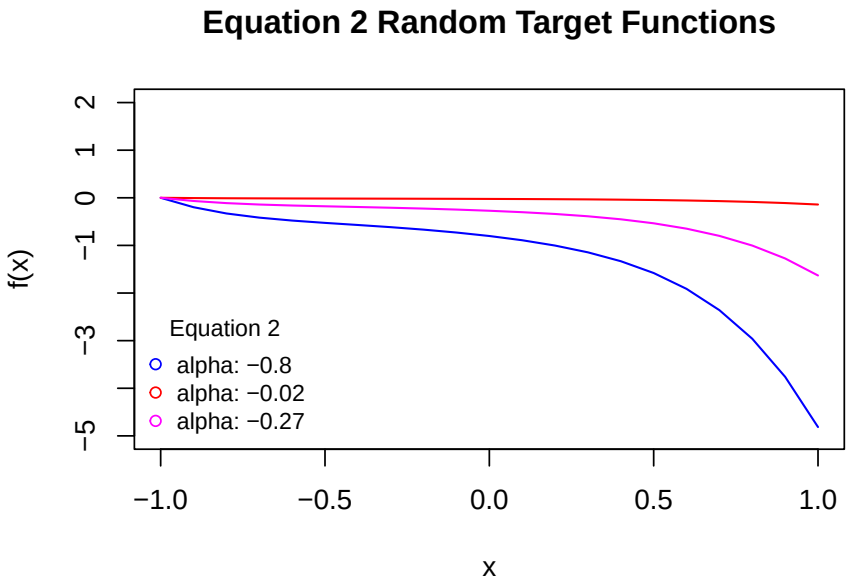
a) Legendre Polynomial Plots

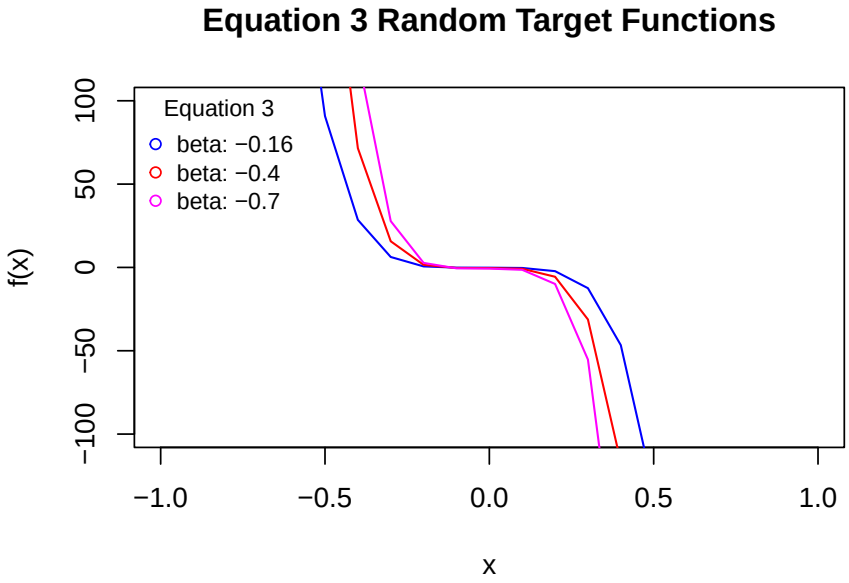


As the order of the polynomial, q, increases, the more nonlinear/complex the$L_q(x)$ gets, albeit slowly.

1. At q=0, $L_q(x)$ devolves to a straight horizontal line $L_q(x)$ =0 over the domain [-1,1].
2. At q=1 $L_q(x)$ starts to have some slope attached, a straight line of the form $y = 2x$
3. At q=2, 4 the non-linear curvature/complexity of $L_q(x)$ starts to become more visible particularly towards the edges of the domain [-1,1; function approximates a parabola/quadratic function
4. At q=3,5 non-linear curvature/complexity of $L_q(x)$ both take the same shape, albeit with different magnitudes of $L_q(x)$ over the same domain.

b) Random Target Functions



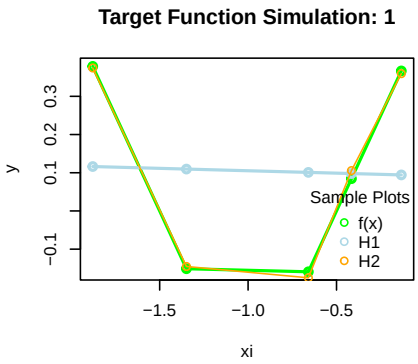


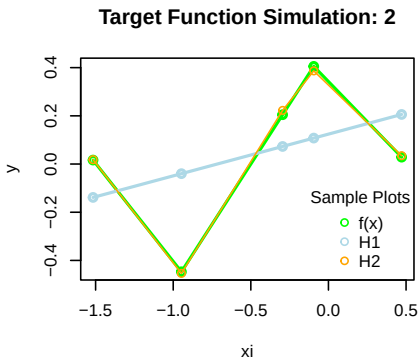
Observations:

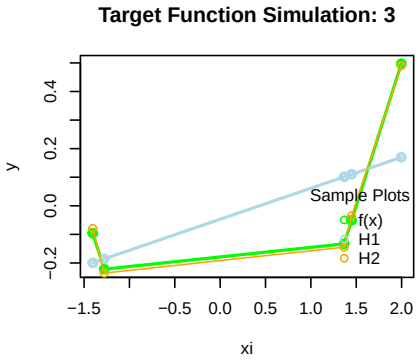
1. $f(x) = \sum_{q=0}^{Q_f} \alpha_q x^q$: the functions only output values from 0 to negative values over the domian [-1,1]
2. $f(x) = \sum_{q=0}^{Q_f} \beta_q L_q(x)$: the functions output both positive and negative y values, however, over the domain [-1,1], the output values are only defined (or real) between 0.5 and -0.5. The appears to be a vertical asymtopte over values outside those bounds.

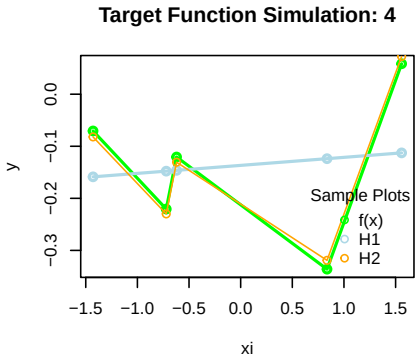
Question 3

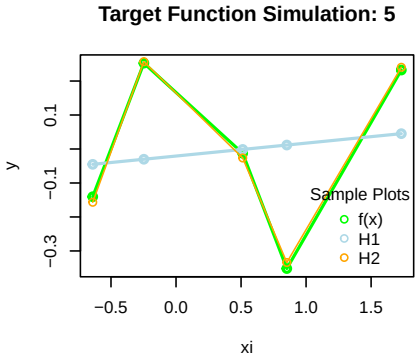
a) Hypothesis Function Simulations









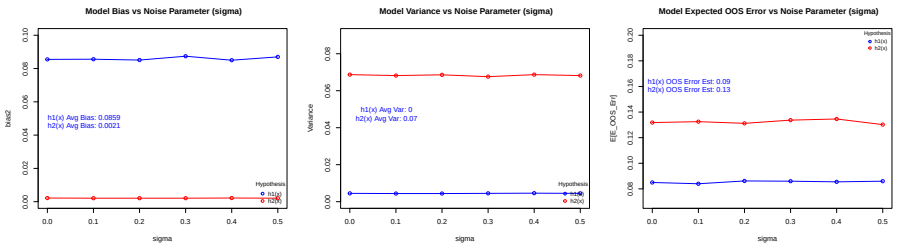


b) A Priori expectation

Given that the underlying data generating process is non-linear, the non-linear hypothesis would be more suited to fit the data's non-linear behavior and overall shape. This visually seems to be the case as well: the H2(x) hypothesis functions follows the shape of the data quite closely.

c) Bias Variance for H1 and H2

	Bias	Variance	Both:Bias+Var	OOS Error	Estimate Differance
h1(x)	0.0854	0.0044	0.0898	0.0841	0.0057
h2(x)	0.0022	0.0703	0.0725	0.1296	0.0571



(a) Bias vs Noise (b) Variance vs Noise (c) E[OOS Error] vs Noise

Figure 1: OOS Error Estimates

e) Interpretation *comeback

- The expected out of sample error as it relates to the bias and variance of a model is given by

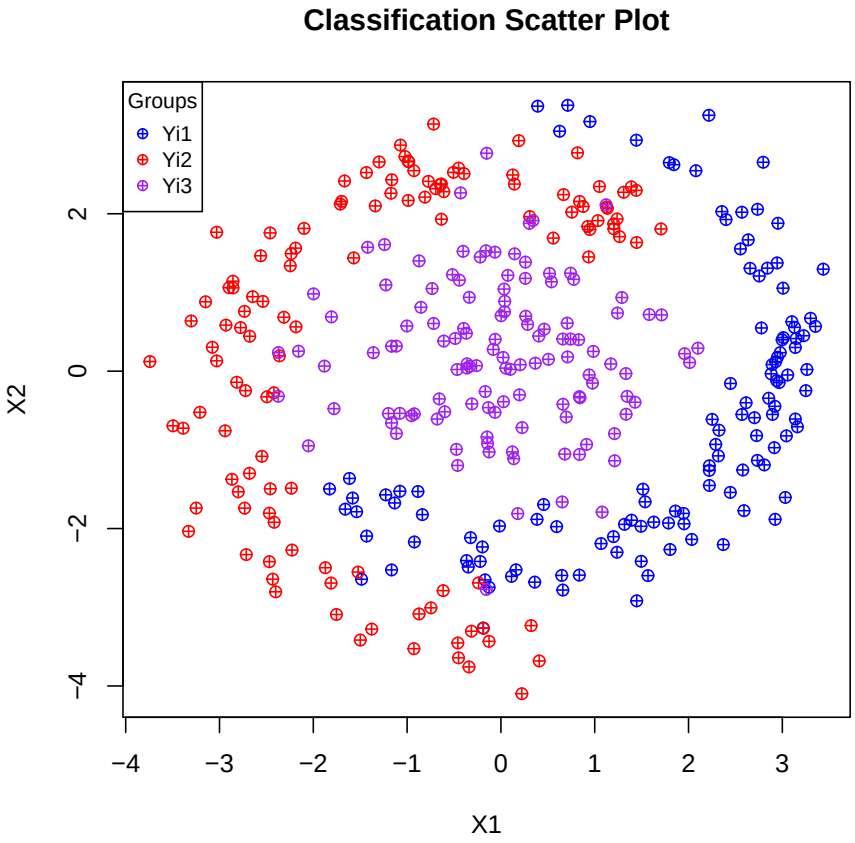
$$E[Err_{oos}] = Bias^2 + Variance + IrreducibleNoise(\varepsilon)$$

- RHS:
  - The linear model , $h1(x)$, has better out-of-sample performance overall averaged across 1000 simulations over varying values of the noise parameter compared to the more complex model, $h2(x)$.
  - The combined
- LHS:

– The linear model also has lower average variance but higher bias across the various noise parameters, this is expected given the that the model is less complex relative to $h2(x)$
– Given the estimated OOS Error, go with the linear model as it is more parsimonious and less likely to overfit the data

Question 4

a) Data Set Plot



Observation: The data does not appear to be linearly separable (in the given 2D feature space) as the data does not seem to have linear boundaries between categories/classes. Instead the different set of responses look spiralled around each other. A non-linear neural network seems well suited to capture the non-linear pattern/structure embedded in the data in for the given feature space.

b)

c) Feed Forward Neural Network

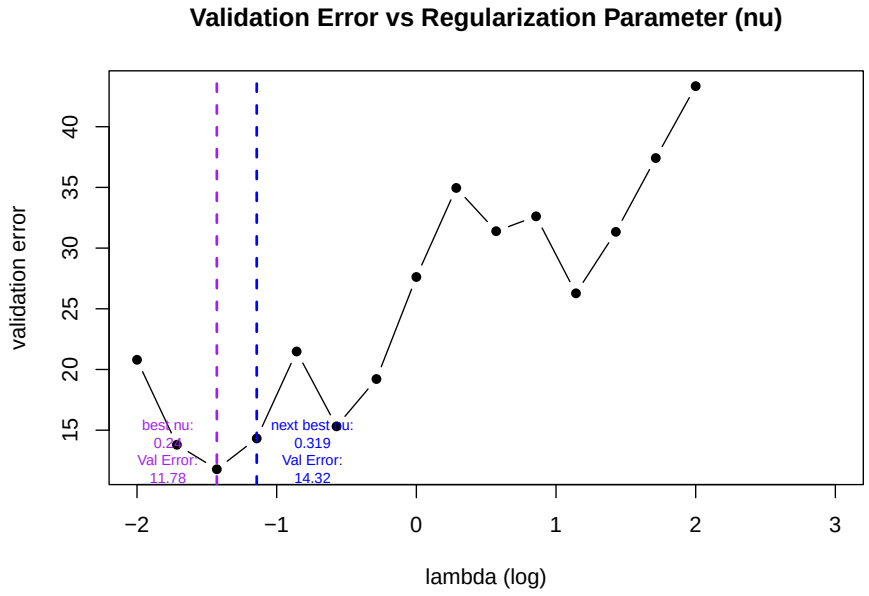
Numerical Pitfalls: The computaion of log(yhat) in the objective function may cause the network to explode/ diverge given the undefined behavior of the

log function at 0 (or values very close to 0), which can hamper the numerical stability of the network's computations.

d) Validation Analysis

Plot the above

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[1] 10
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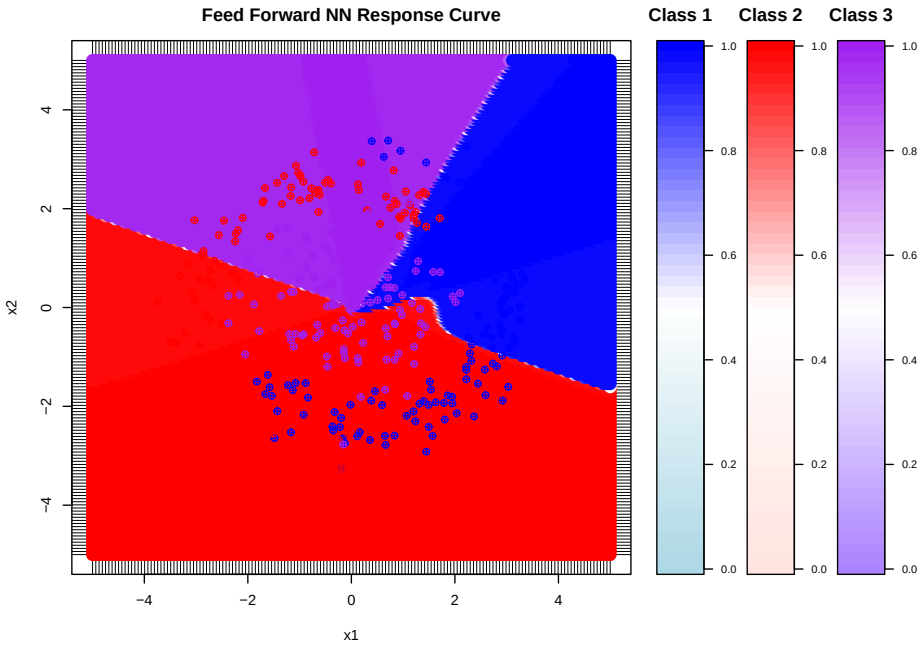


Model Parameters:

The lowest validation error is generated at $\lambda_{min} = 2.356$ with an associated validation making for a relatively stricter and simpler model. The next best error to the right of this minimum is obtained at $\lambda_{1SE} = 3.136$, offering stricter control over the complexity of the network and thus more parsimonius model, decreasing the likelihood of overfitting to the training data.

d) Response Curve (?)

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The model produces three distinct regions over the feature space that correspond to the respective probability assignments of each class:

- Class 1 labels were mostly misclassified by the model, with the predicted region for that class for the model containing few of the actual observations for that label. The model also quite confident in its predictions for that region with high probabilities assignments of lattice points assigned to the class.
- For observations with values between [-4,0] for X1 and between [-4,~0.4], the model mostly predicts Class 2 with relatively high probabilities (close to 1).
- Class 3 probability predictions region dominates the feature space. The majority of observations with values $>$ -2.24 along the x2 dimension are predicted as belonging to Class 3, with increased probabilities for this class at higher values of x2.